

StylistAI — 15-Minute Last-Minute Cheat Sheet

Quick Memory Refresh for Technical Interview

ELEVATOR PITCH (30 SECONDS):

"StylistAI is a voice-enabled AI appointment booking assistant for hair salons. Built with React 19, Node/Express, Groq LLaMA 3.3 70B, Supabase PostgreSQL, and Google Calendar API. It parses natural language booking requests, enforces stylist skill matching, evaluates real-time dual-layer slot availability (Supabase + Google Calendar), creates calendar invites with client reminders, and dispatches EmailJS receipts."

CORE ARCHITECTURE & STACK SNAPSHOT

Layer	Technology	Key File	Core Function
Frontend	React 19 + Vite	client/src/pages/ChatPage.jsx	Neumorphic Bento UI, state management, Web Speech API voice TTS.
Backend API	Node.js / Express	server/src/index.js	CORS, Express JSON (10kb), routing, global error handler.
AI Intent Engine	Groq SDK (LLaMA 3.3 70B)	server/src/ai/groq.js	Sub-300ms inference, JSON mode extraction, Zod validation, 3-tier cascade.
Availability Engine	Node.js Custom Engine	server/src/services/availabilityService.js	30-min slot math (9AM-6PM), dual-layer conflict checking (Supabase + GCal).
Database	Supabase PostgreSQL	server/src/db/migration_salon.sql	Services, employees, employee_services junction table, unique slot index.
Calendar API	Google Calendar v3 (OAuth)	server/src/services/calendarService.js	OAuth 2.0 token storage, busy intervals, events with 24h & 30m reminders.
Email Dispatch	EmailJS Browser SDK	client/src/services/emailService.js	Client-side HTML confirmation receipts dispatches upon booking.
Admin Portal	React 19 Dashboard	client/src/pages/AdminPage.jsx	Passphrase auth gate, live KPI grid, appointment controls, AI telemetry.

TOP 10 INTERVIEW MUST-KNOWS

1. **AI vs Backend Split:** AI ONLY parses language into JSON; backend ENFORCES availability & business rules.

2. **Why Groq?** ~280ms inference latency vs 2-3s on standard cloud GPUs makes conversational chat seamless.

3. **Model Cascade:** groq/compound -> llama-3.1-8b-instant -> gemma2-9b-it.

4. **Double-Booking Prevention:** Supabase partial index appointments_employee_slot_unique on (employee_id, date, time) WHERE status != 'cancelled'.

5. **Dual Availability Check:** Combines Supabase booked slots AND Google Calendar busy intervals in isSlotAvailable().

6. **Stylist Capability Match:** Checked via employee_services junction table before presenting time slots.

7. **No Stylist Preference:** findBestAvailable() ranks open slots across all capable stylists near preferred time.
8. **Google OAuth 2.0:** Salon owner authenticates via /api/calendar/auth; tokens stored in tokens/google-tokens.json.
9. **Voice Speech:** 100% client-side Web Speech API (SpeechRecognition + SpeechSynthesis).
10. **Non-Blocking External Services:** GCal or EmailJS failures log warnings but DO NOT block Supabase booking insertion.

TOP 5 TRAP QUESTIONS & INSTANT ANSWERS

- **Q: Does the AI check database availability?**

A: No! The AI only extracts intent JSON. Express backend queries Supabase and Google Calendar.

- **Q: What happens if two users book Sarah at 4 PM at the exact same second?**

A: Pre-check allows both, but Supabase enforces appointments_employee_slot_unique. The second insert fails with error 23505 and prompts user for a new time.

- **Q: Why use OAuth 2.0 instead of a Service Account?**

A: OAuth 2.0 lets the salon owner connect their real primary Google Calendar in 1 click without complex Google Workspace delegation.

- **Q: Why Zod if Groq already outputs JSON?**

A: JSON mode ensures valid syntax, but Zod guarantees exact field types and enums required by the backend.

- **Q: Can Sarah and David both have appointments at 4 PM?**

A: Yes! Uniqueness is per employee slot (employee_id, date, time), not global slot.